The Discrete Fourier Transform (DFT):

- The DFT is a computational tool used to analyze the frequency content of *finite-length* discrete-time signals.
- Our DFT discussion will include:
  1. definitions of the DFT and IDFT,
  2. interpretations of the DFT:
     (a) sampled DTFT,
     (b) discrete Fourier series,
  3. properties of the DFT, including circular convolution,
  4. use of DFT for spectral analysis,
  5. fast computation of the DFT via the FFT,
  6. matrix/vector formulations.

$\left\{e^{-j\frac{2\pi k}{N}n}\right\}_{n=0}^{N-1} \cdot \left\{x[n]\right\}_{n=0}^{N-1}$

DFT definitions:

- For an N-length signal $\{x[n]\}_{n=0}^{N-1}$, the N-point DFT is

analysis

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, \qquad k = 0 \ldots N-1$$

and the corresponding N-point IDFT is

synthesis

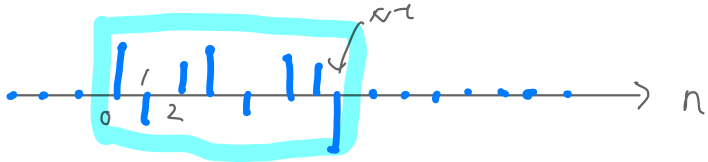
$$\boxed{x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn},} \qquad n = 0 \ldots N-1.$$

- Note:
  - the signal has finite duration N,
  - the frequency domain is discrete and length-N.

$$[W]_{k,n} = e^{-j\frac{2\pi}{N}kn}$$

$$W^H W = N I_N$$

$$W^{-1} = \frac{1}{N} W^H$$



a DT signal supported on $\{0, 1, 2, \cdots, N-1\}$

DFT Interpretation # 1 — sampled DTFT:

Consider the finite-length signal $\{x[n]\}_{n=0}^{N-1}$. Defining $\{x[n]\}_{n=-\infty}^{\infty}$ via zero extension, the DTFT becomes

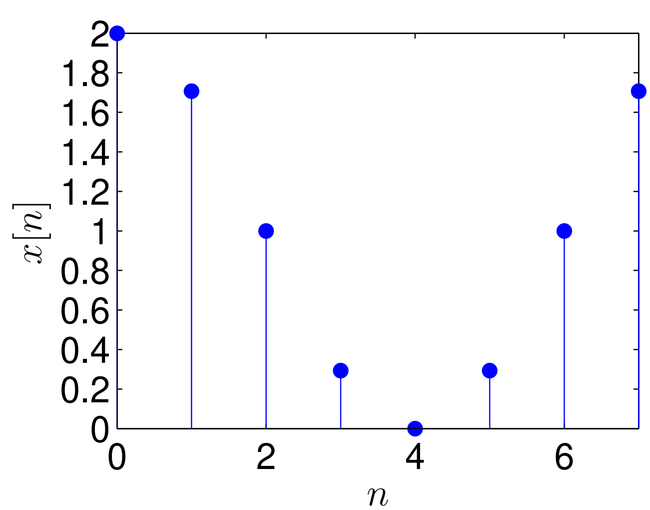
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=0}^{N-1} x[n]e^{-j\omega n}.$$

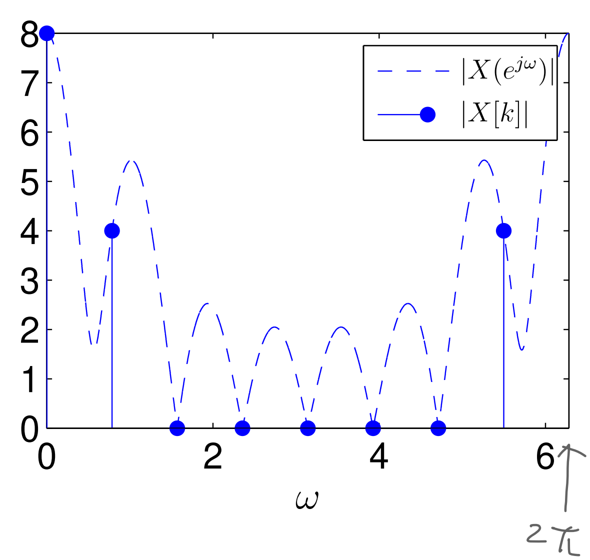
Meanwhile, the N-DFT of the non-zero samples is

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn} = X(e^{j\omega})\Big|_{\omega=\frac{2\pi}{N}k}$$

for $k = 0 \ldots N-1$.

Thus, the N-DFT returns $\frac{2\pi}{N}$-spaced samples of the (zero-extended) DTFT. Example for $N = 8$:





Notice that we've uncovered a new concept: *sampling in the frequency domain!*

inverse DTFT

$$x[n] = \frac{1}{2\pi}\int_0^{2\pi} X(e^{j\omega})e^{j\omega n}\,d\omega$$

We need to know $X(e^{j\omega})$ for all $0 \leq \omega < 2\pi$

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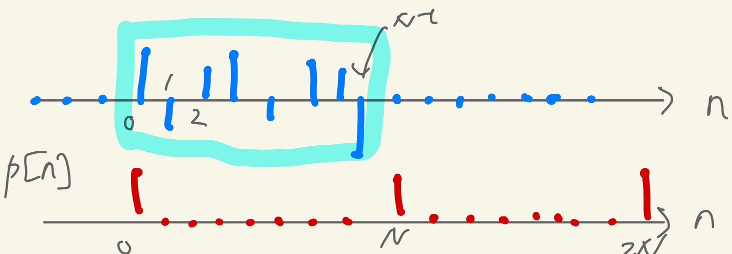
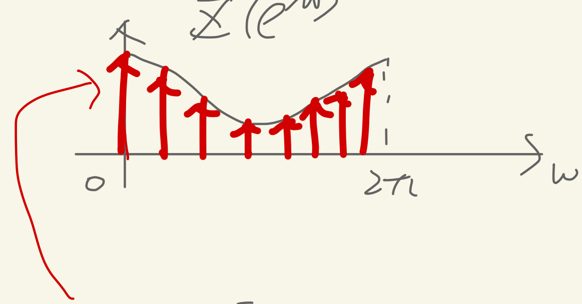
inv DFT $\quad x[n] = \frac{1}{N}\sum_{k=0}^{N-1} \underbrace{X[k]}_{X(e^{j\omega})\big|_{\omega=\frac{2\pi k}{N}}} e^{j\frac{2\pi k}{N}n}$

$$p[n] = \sum_{m=-\infty}^{\infty} \delta[n+Nm] = \delta[\langle n\rangle_N]$$

$$P(e^{j\omega}) = \frac{2\pi}{N}\sum_{l=-\infty}^{\infty}\delta\left(\omega - \frac{2\pi}{N}l\right)$$





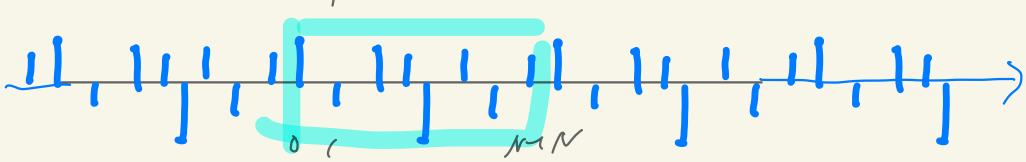
$x[n] * p[n] \quad \longleftrightarrow \quad X(e^{j\omega})\,P(e^{j\omega})$

$x[n] * \delta[n] = x[n]$

$x[n] * \delta[n-N] = x[n-N]$

$$x[n] * \sum_m \delta[n+Nm] = \sum_m x[n+Nm]$$

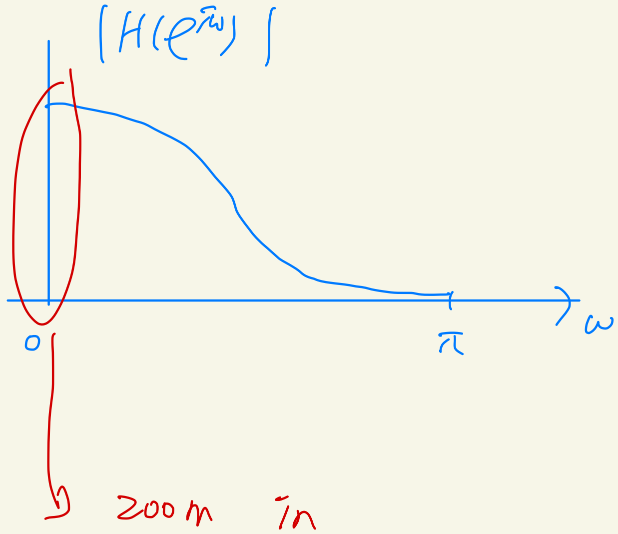
$x[n] * p[n]$



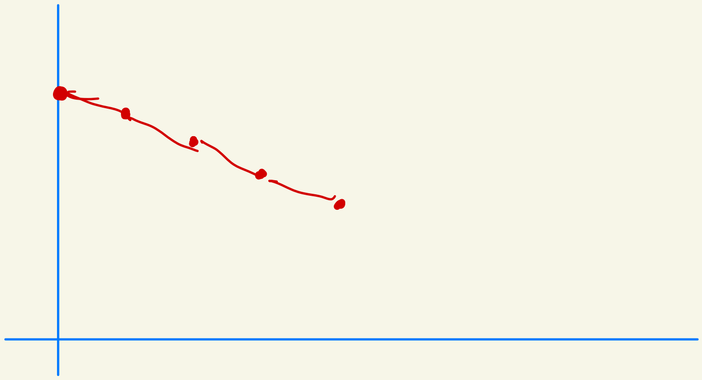
$$\frac{1}{2\pi}\int_0^{2\pi} \underbrace{\frac{2\pi}{N}\sum_{k=0}^{N-1} \delta\left(\omega - \frac{2\pi k}{N}\right)}_{P(e^{j\omega}) \text{ for } 0 \le \omega < 2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

$$= \frac{1}{N}\sum_{k=0}^{N-1} \underbrace{\int_0^{2\pi} \delta\left(\omega - \frac{2\pi k}{N}\right) X(e^{j\omega}) e^{j\omega n}\, d\omega}$$

$$\underbrace{X(e^{j\omega})\Big|_{\omega = \frac{2\pi k}{N}}}_{X[k]} e^{j\frac{2\pi k n}{N}}$$



$|H(e^{j\omega})|$
0
π
ω
zoom in



The DFT and zero-padding:

Again consider a finite-length $\{x[n]\}_{n=0}^{N-1}$ and $\{x[n]\}_{n=-\infty}^{\infty}$ defined via zero-extension.

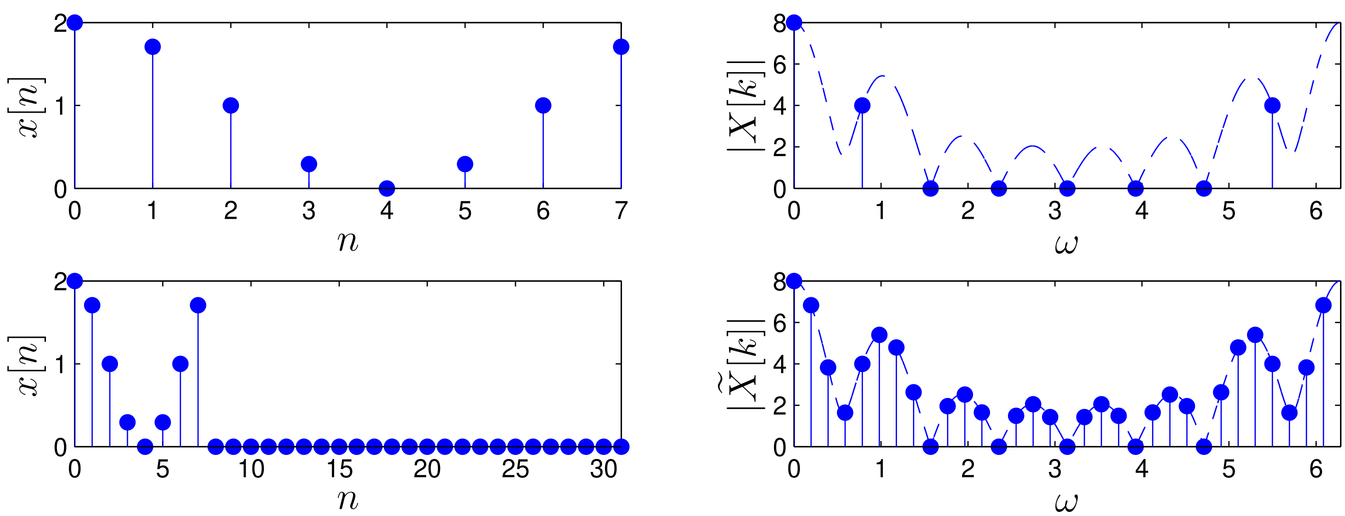
But now take the M-length DFT, where $M > N$:

$$\begin{aligned} \widetilde{X}[k] &= \sum_{n=0}^{M-1} x[n] e^{-j\frac{2\pi}{M}kn} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\frac{2\pi}{M}kn} \\ &= X(e^{j\omega})\big|_{\omega=\frac{2\pi}{M}k}. \end{aligned}$$

The M-length DFT samples the DTFT at the interval $\frac{2\pi}{M}$.

The process of appending zeros to a finite-length signal is called *zero padding*.

Thus, zero-padding increases the rate at which the DFT samples the DTFT!



$\sum_{k=-\infty}^{\infty} \delta(t-Pk) \xleftrightarrow{FT} \frac{2\pi}{P} \sum_{k} \delta(\Omega - \frac{2\pi}{P}k)$

$x(t) = \tilde{x}(t) * \sum_k \delta(t-Pk)$

DFT Interpretation #2 — discrete-time FS:

Say $\tilde{x}(t)$ is periodic with P-second period, and $x(t)$ contains only the first period of $\tilde{x}(t)$. Then, from Fourier series,

$$x(t) = \frac{1}{P} \sum_{k=-\infty}^{\infty} P X_k e^{j\frac{2\pi}{P}kt}, \quad t \in [0, P)$$

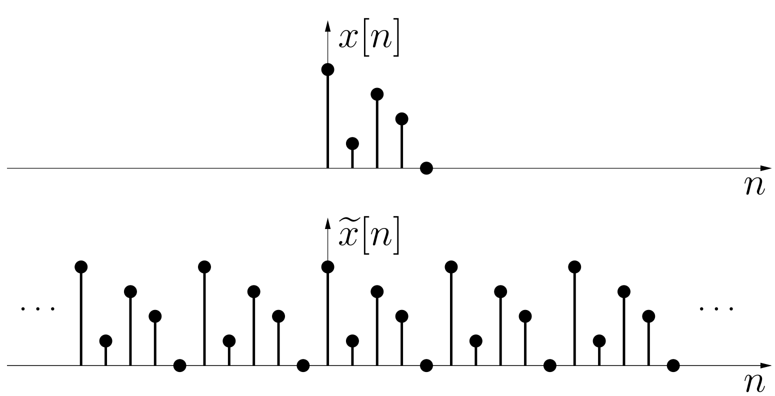
$$P X_k = \int_0^P x(t) e^{-j\frac{2\pi}{P}kt} dt = X_c(j\Omega)\Big|_{\Omega=\frac{2\pi}{P}k}$$

Say $\tilde{x}[n]$ is periodic with N-sample period, and $x[n]$ contains only the first period of $\tilde{x}[n]$. Then, from the DFT,

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}, \quad n = 0 \ldots N-1$$

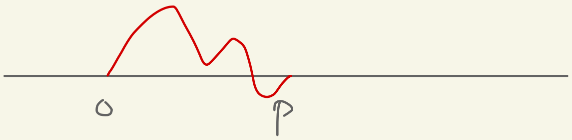
$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} = X(e^{j\omega})\Big|_{\omega=\frac{2\pi}{N}k}$$

The similarity suggests that one can interpret the DFT coefs $\{X[k]\}_{k=0}^{N-1}$ as the coefs of a "discrete-time Fourier series."

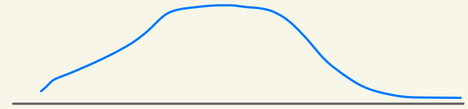


Note: zero-extension $\rightsquigarrow$ DTFT; periodic extension $\rightsquigarrow$ FS.

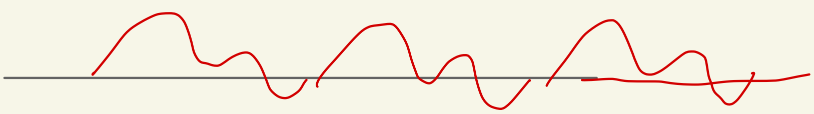
$\tilde{x}(t)$

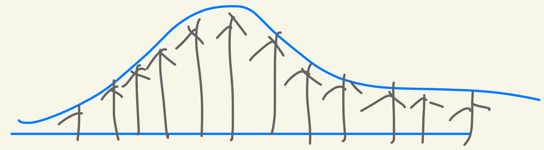


$X(j\Omega)$



$$x(t) = \tilde{x}(t) * \sum_{k} \delta(t-kP)$$





$$\begin{bmatrix} x[0]\\ x[1]\\ \vdots \\ x[N-1]\end{bmatrix} = \boxed{\underbrace{\left[\frac{1}{N} e^{j\frac{2\pi}{N}kn}\right]_{n,k}}_{N\times N}} \begin{bmatrix} X[0]\\ X[1]\\ \vdots\\ X[N-1]\end{bmatrix} \qquad [a_{ij}]_{i,j} = \begin{bmatrix} a_{11} & \cdots & a_{1N}\\ \vdots & & \\ a_{N1} & & a_{NN}\end{bmatrix}$$

DFT interpretation #3 — basis expansion:

Recall from IDFT that $\quad x[n] = \displaystyle\sum_{k=0}^{N-1} X[k] \underbrace{\tfrac{1}{N} e^{j\frac{2\pi}{N}kn}}_{\triangleq\, b_k[n]}.$

IDFT

Thus, any $\{x[n]\}_{n=0}^{N-1}$ can be expressed as a weighted sum of N orthogonal *basis waveforms* $\{b_k[n]\}_{n=0}^{N-1}$.

$$\begin{bmatrix} b_k[0]\\ \vdots \\ b_k[N-1]\end{bmatrix}$$

is

kth column

$$\sum_{n=0}^{N-1} b_k^*[n] b_l[n] = \frac{1}{N} \cdot \frac{1}{N} \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(l-k)n} = \frac{1}{N}\delta[l-k]$$

These act as elementary "building blocks" for signals.

Example for $N = 8$: